🛒 Zepto Inventory Analysis — SQL + Inferential Statistics

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📂 Folder Structure

```
zepto-inventory-analysis/
|
```

```
├── data/
│   └── zepto.csv                  # Raw dataset
│
├── sql/
│   └── sql_1_project.sql          # All SQL queries with business insights
│
├── notebooks/
│   └── stats_analysis.ipynb       # Inferential statistics in Python
│
└── README.md
```

🌟 STAR Framework — Project Overview

📌 Situation

Zepto is a quick-commerce platform that promises 10-minute grocery delivery across India. In this business model, **inventory management is everything** — a product going out of stock means instant revenue loss, and incorrect pricing or unplanned discounting directly eats into margins.

The dataset contains **product-level inventory data** across 7 categories including Fruits & Vegetables, Dairy, Beverages, Munchies, Packaged Food, Cooking Essentials, and Bread & Batter — with attributes like MRP, discount %, available quantity, weight, selling price, and out-of-stock status.

🎯 Task

The goal was to go beyond surface-level querying and answer **3 core business questions** using both SQL analytics and statistical hypothesis testing:

1. **Availability Problem** — Which products and categories are going out of stock, and what drives stockouts — is it pricing or discounting?
2. **Pricing & Discount Problem** — Are discounts applied strategically across categories, or is there no clear pattern?
3. **Revenue Problem** — Which categories generate the most estimated revenue, and which are underperforming?

The key challenge was to determine whether the patterns observed in the data are **statistically significant** or just random noise.

⚙️ Action

🔷 SQL Analysis (MySQL Workbench)

#	Query	Business Purpose
Q1	Top 10 products by discount %	Identify potential margin or low-selling products
Q2	High MRP products that are OOS	Restocking priority for premium items
Q3	Estimated revenue per category	Guide marketing campaigns
Q4	High MRP + low discount products	Identify overpriced, low-offer risk products
Q5	Top categories by avg discount %	Evaluate discount strategy per category
Q6	Price per gram for products > 100g	Find best value-for-money products
Q7	Weight-based product classification	Help store managers sort inventory efficiently
Q8	Total inventory weight per category	Warehouse space planning
Q9	Top 3 discounted products per category	Identify brand-level discount leaders (Window Function)
Q10	Revenue ranking per category	Category-level performance benchmarking (Window Function)

🔷 Inferential Statistics (Python — Jupyter Notebook)

Test 1 — Independent T-Test: MRP vs Out-of-Stock Status

- **H0:** Mean MRP is the same for in-stock and out-of-stock products
- **Tool:** `scipy.stats.ttest_ind`

```python
in_stock  = df[df['outOfStcok'] == 'False']['mrp'].dropna()
out_stock = df[df['outOfStcok'] == 'True']['mrp'].dropna()
t_stat, p_val = stats.ttest_ind(in_stock, out_stock)
```

Metric	Value
In-Stock Mean MRP	₹151.26
Out-of-Stock Mean MRP	₹90.78
T-Statistic	4.4727
P-Value	0.0000

✅ **Rejected H0** — Statistically significant difference found ($p < 0.05$)

Test 2 — Point-Biserial Correlation: Discount % vs Out-of-Stock Status

- **H0:** No correlation between discount % and OOS status
- **Tool:** `scipy.stats.pointbiserialr`
- **Why Point-Biserial?** — One variable is continuous (discountPercent), other is binary (OOS True/False)

```python
corr, p_val = pointbiserialr(clean['oos_binary'], clean['discountPercent'])
```

Metric	Value
Correlation Coefficient	-0.0757
P-Value	0.0020

✅ **Significant but weak negative correlation** found ($p < 0.05$)

📊 **Result**

Finding 1 — MRP drives stockouts more than discounts:

> Low-to-mid priced products (avg ₹90) go out of stock significantly more than premium products (avg ₹151). This suggests higher demand velocity for affordable products — a clear restocking priority signal for Zepto's procurement team.

Finding 2 — Discount % is not a reliable stockout predictor:

> While a statistically significant negative correlation exists between discount % and OOS status ($r = -0.07$, $p < 0.05$), the effect size is too weak to conclude that discounting directly controls stockout behavior. Pricing and supply chain factors are stronger drivers.

Finding 3 — Revenue concentration:

> Category-wise revenue analysis revealed that certain categories disproportionately contribute to estimated revenue — useful for the marketing team to prioritize ad spend and promotions.

Overall Conclusion:

> *Pricing (MRP) has a stronger influence on stockout behavior than discounting. Affordable products face higher stockout risk, while discount % alone is not a reliable predictor of inventory availability.*

🛠️ Tech Stack

Tool	Usage
MySQL	Data storage, querying, cleaning
Python (Pandas)	Data manipulation
SciPy	Hypothesis testing
Statsmodels	Z-Test, post-hoc tests
Jupyter Notebook	Analysis environment

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